Student id → 202618052

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LAB-07 (DBMS)

1) Draw Relational schema

* Relation1 : Book

ISBN	Title	Amount	Currency	PubID
				P.K.

* Relation2 : BookAuthorName

Book	AuthorName
F.K.	

* Relation3 : Publisher

PubID	PublisherName

* Relation4 : Reprint

Re-ISBN	Amount	Currency	PubID	ISBN
			P.K	F.K

2) Minimal Set of Functiondependencies :-

- Relation1 :

{
ISBN → Title,
ISBN → Amount,
ISBN → Currency,
ISBN → PubID
}

- Relation 3 :

{ PubID → PublisherName }

- Relation 2 :

{ ϕ }

- Relation 4 :

{
Re-ISBN → Amount,
Re-ISBN → Currency,
Re-ISBN → PubID,
Re-ISBN → ISBN
}

→ ∴ All are minimal set.

3) Projected FD :

Relation	Project FD's
1	ISBN → Table, Amount, Currency, PubID
2	None
3	PubID → PublisherName
4	Re-ISBN → Amount, Currency, PubID, ISBN.

(4) Key:

Relation	Key	Closure
1	ISBN	{ISBN}+ = {ISBN, Table, Amount, Currency, PubID}
2	(ISBN, AuthorName)	None
3	PubID	{PubID}+ = {PubID, PublisherName}
4	Re-ISBN	{Re-ISBN}+ = {Re-ISBN, Amount, Currency, PubID, ISBN}

→ ~~Hence~~ Hence, by this we can verify, the key we have taken, will give us closure of that relation.

(5) ER-Model is BCNF :-

→ From minimal set of all relation:

Relation 1: On left side the key is there, there is no non-key on left side, hence BCNF.

Relation 2: As it does not contain non-trivial, so it is in BCNF.

Relation 3: Same logic applied as in relation 1, so it is in BCNF.

Relation 4: Same logic applied as in relation 1, so it is in BCNF.

∴ Hence all relation are in BCNF.

∴ So the given ER Model is in BCNF.